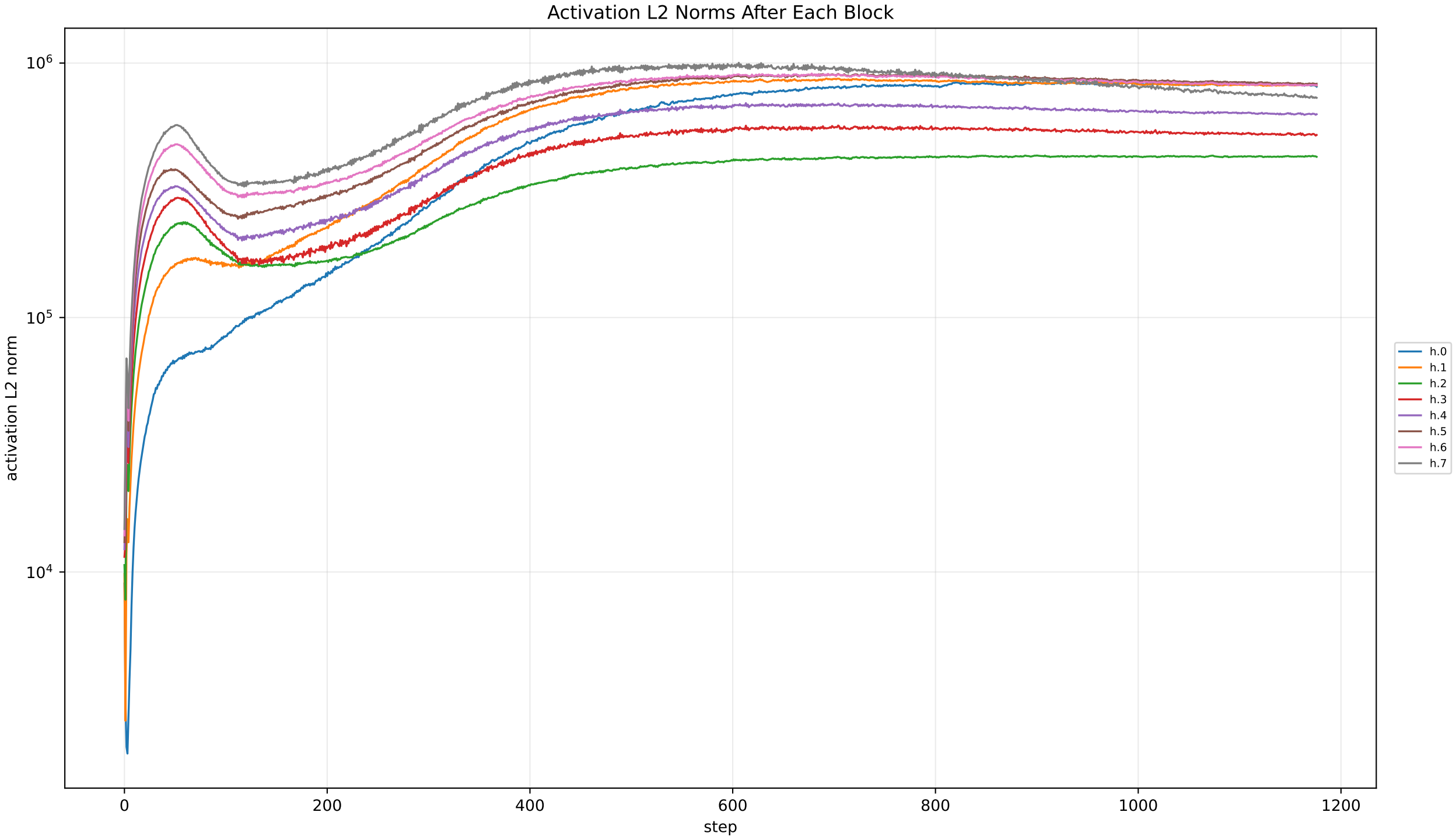
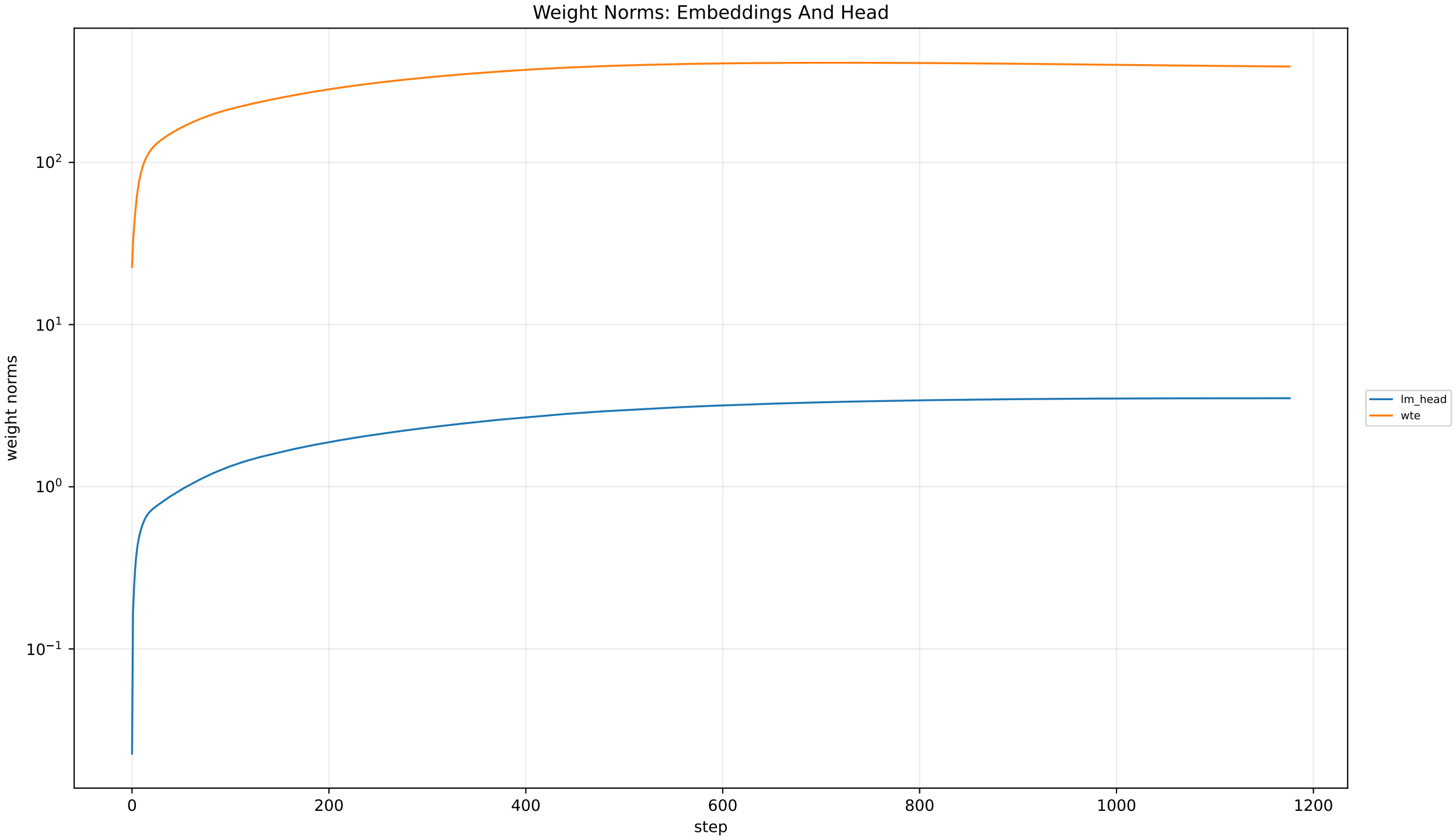
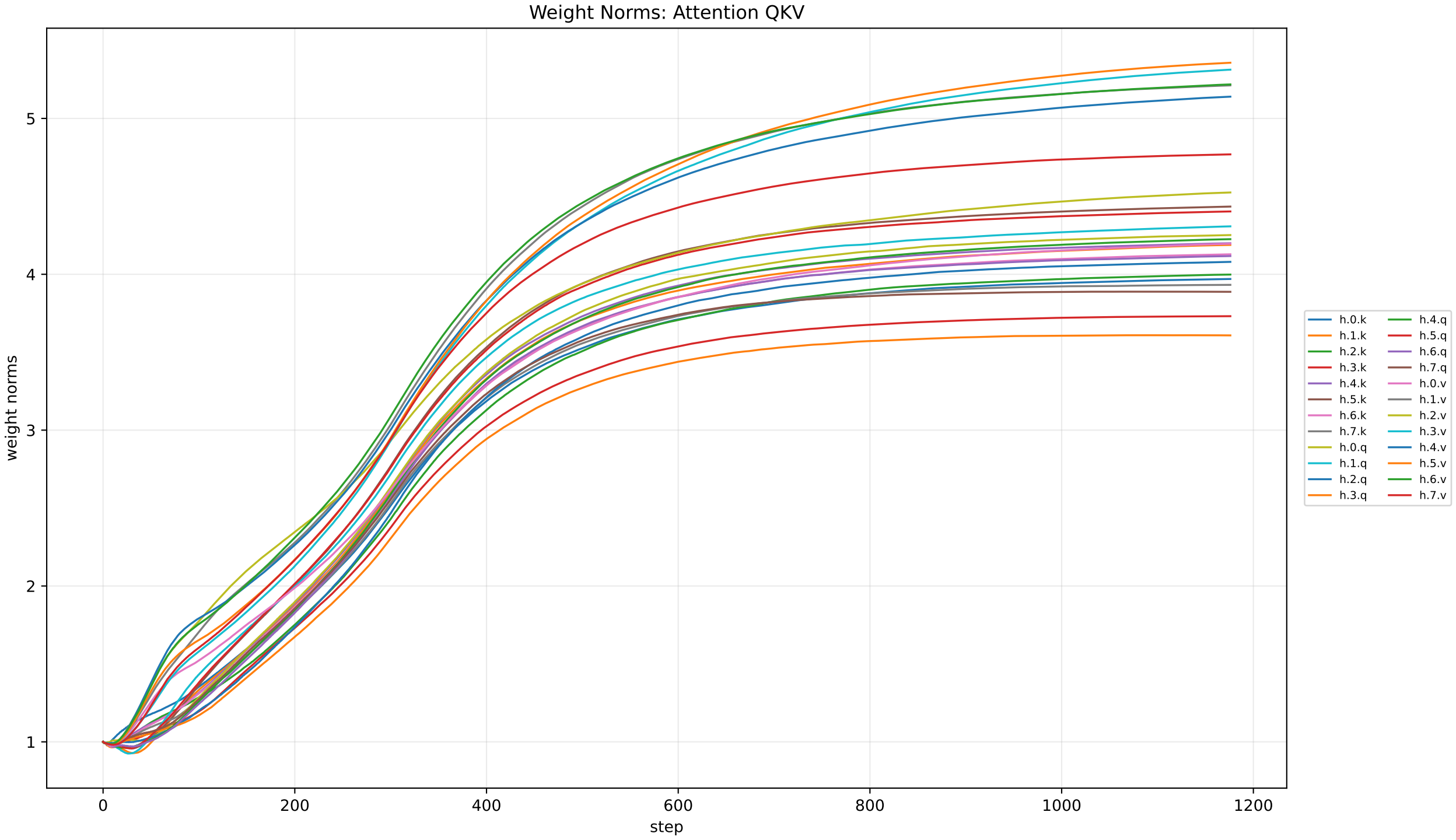
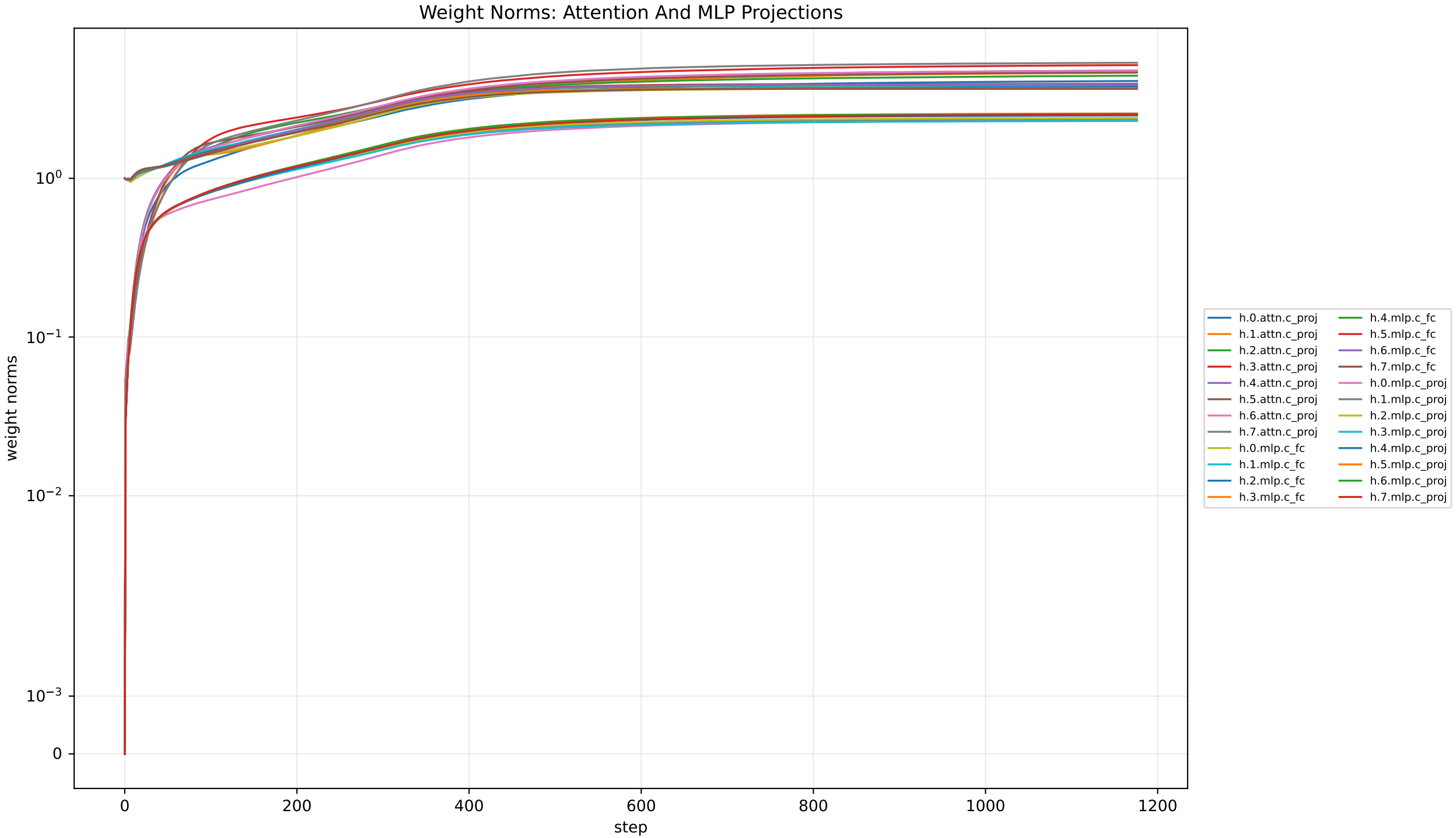
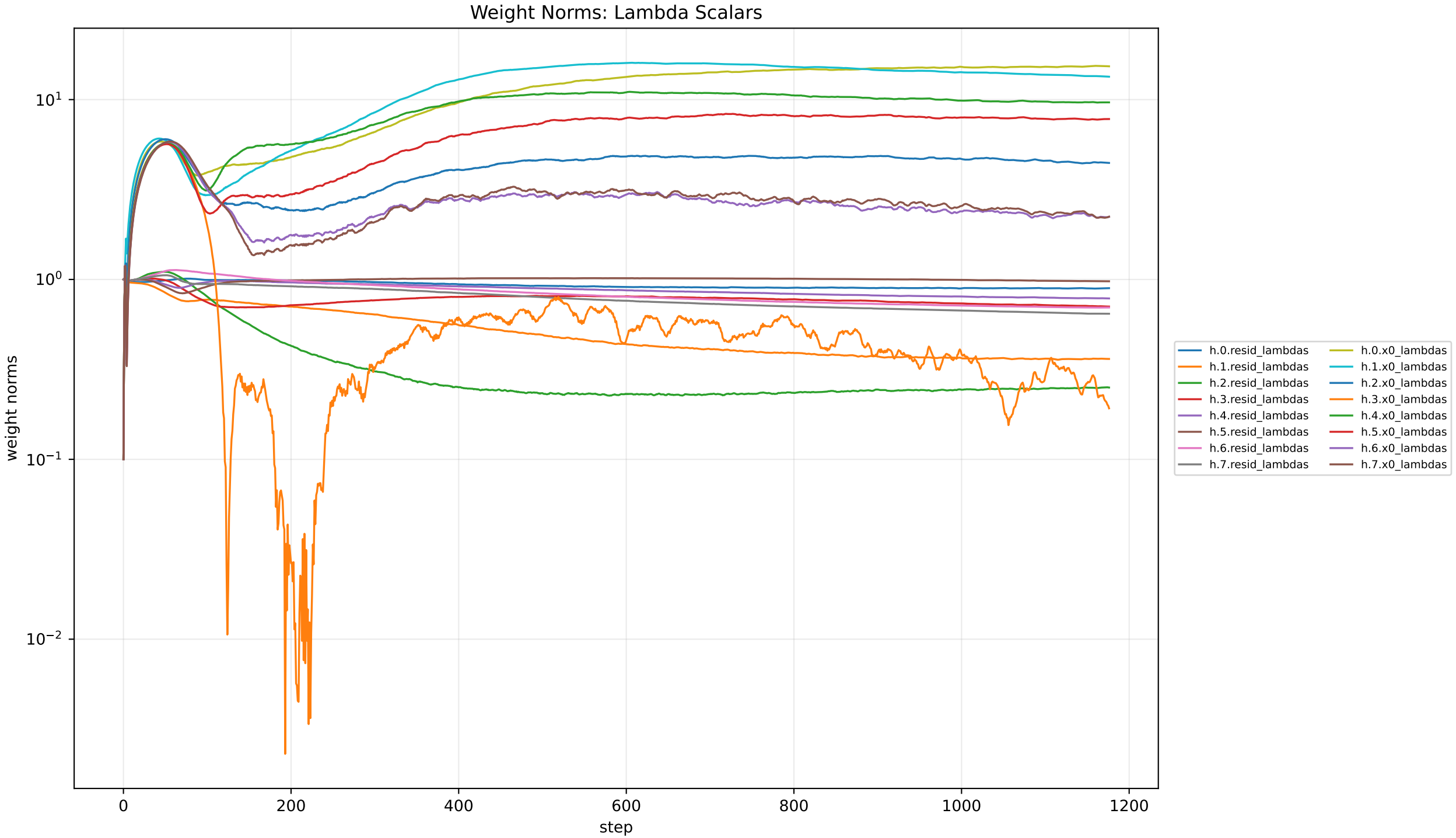
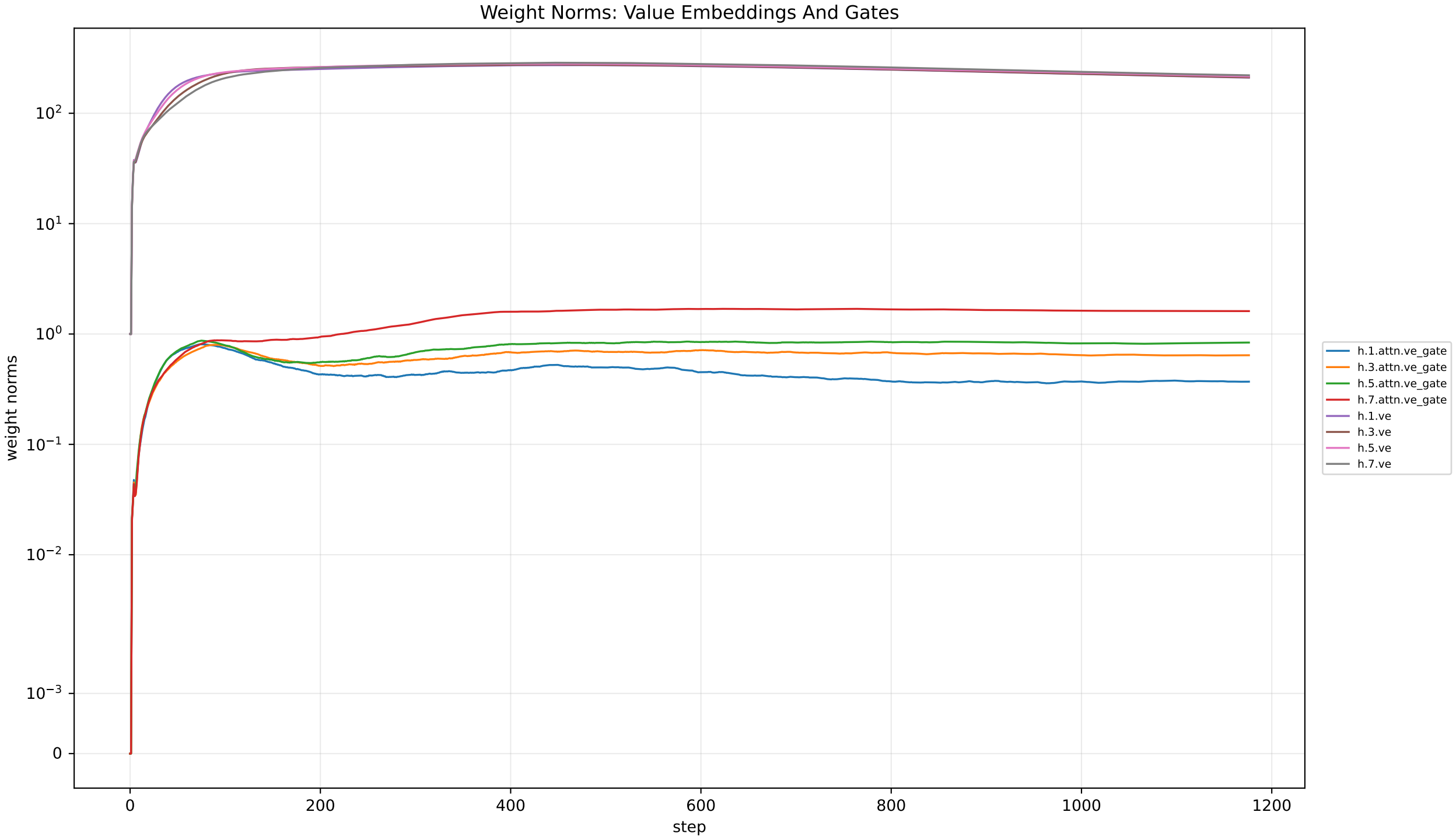

Activation L2 Norms After Each Block
activation L2 norm
step
10^6
10^5
10^4
0
200
400
600
800
1000
1200
h.0
h.1
h.2
h.3
h.4
h.5
h.6
h.7

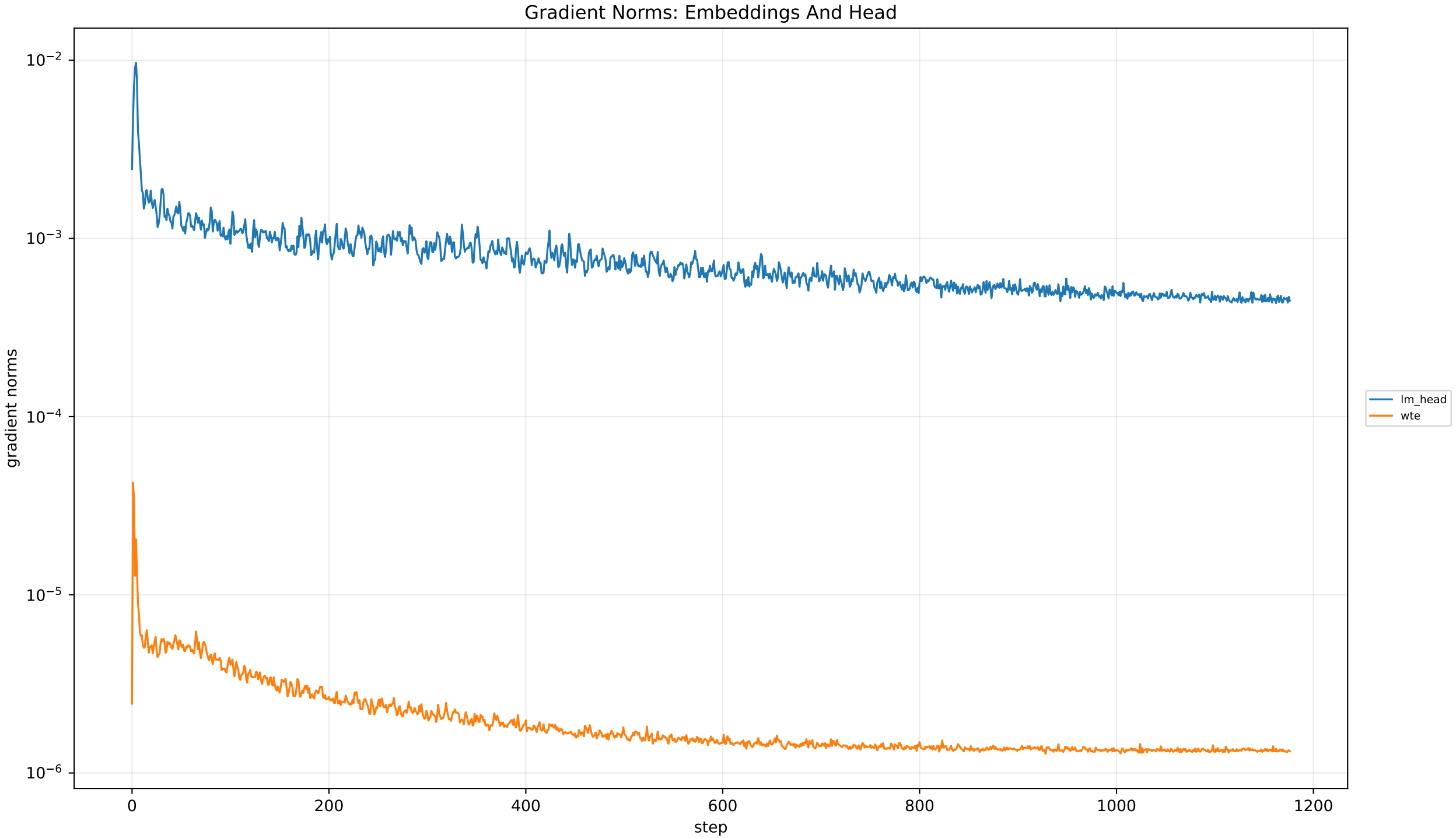

Weight Norms: Embeddings And Head
weight norms
step
10^{2}
10^{1}
10^{0}
10^{-1}
0
200
400
600
800
1000
1200
lm_head
wte

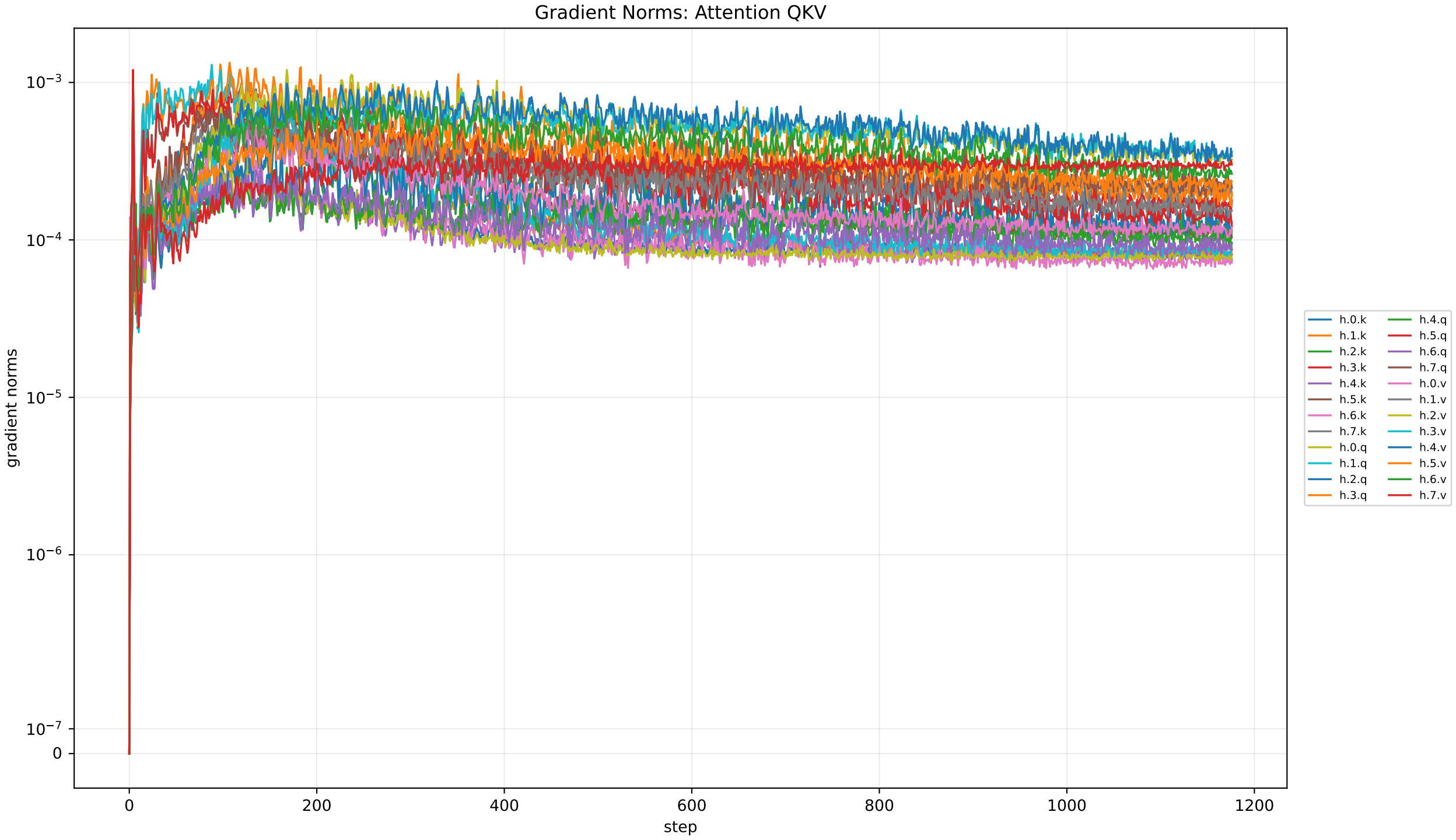

Weight Norms: Attention QKV
weight norms
step
1
2
3
4
5
0
200
400
600
800
1000
1200
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v

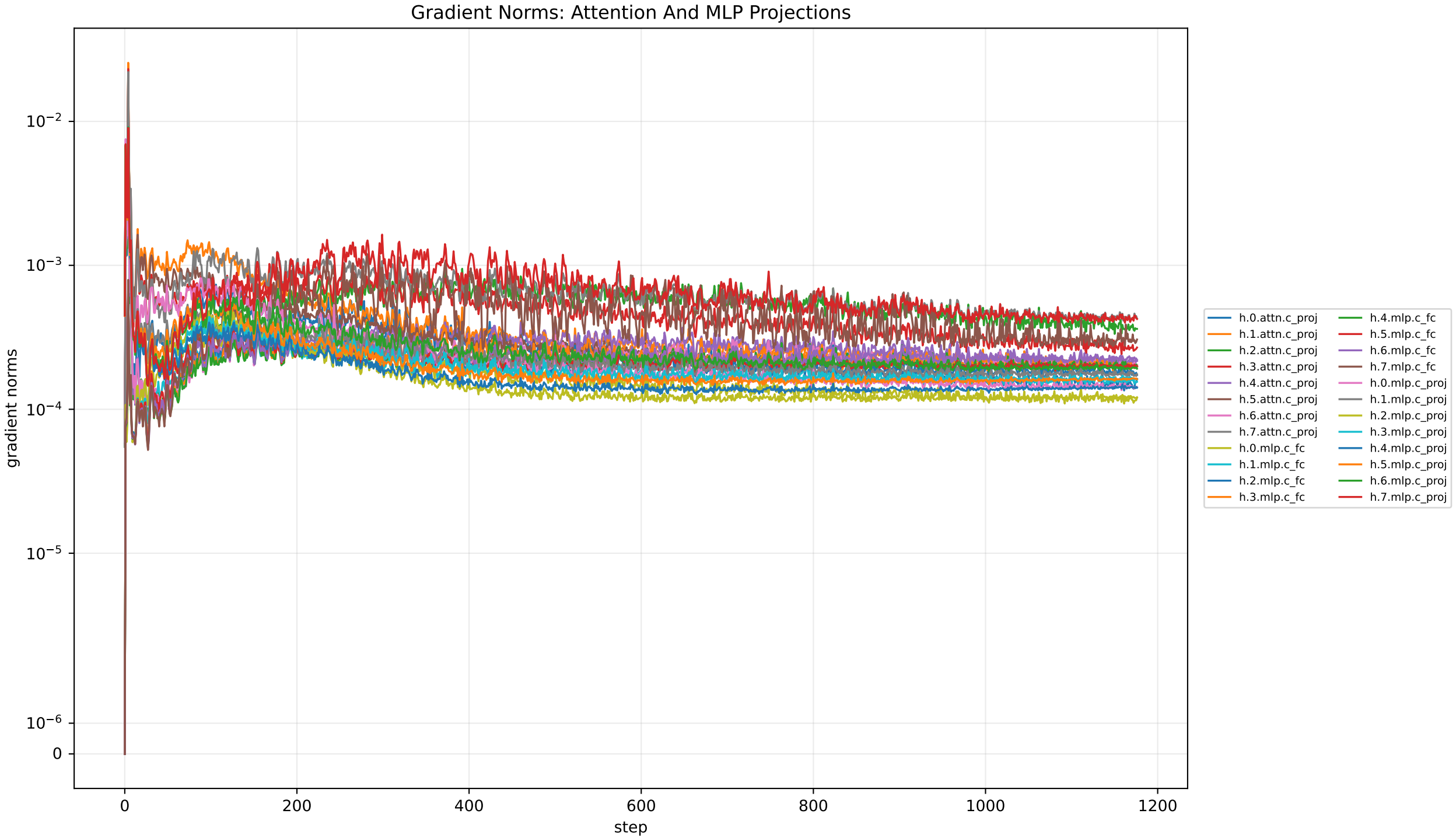

Weight Norms: Attention And MLP Projections
weight norms
step
10^0
10^{-1}
10^{-2}
10^{-3}
0
0
200
400
600
800
1000
1200
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj

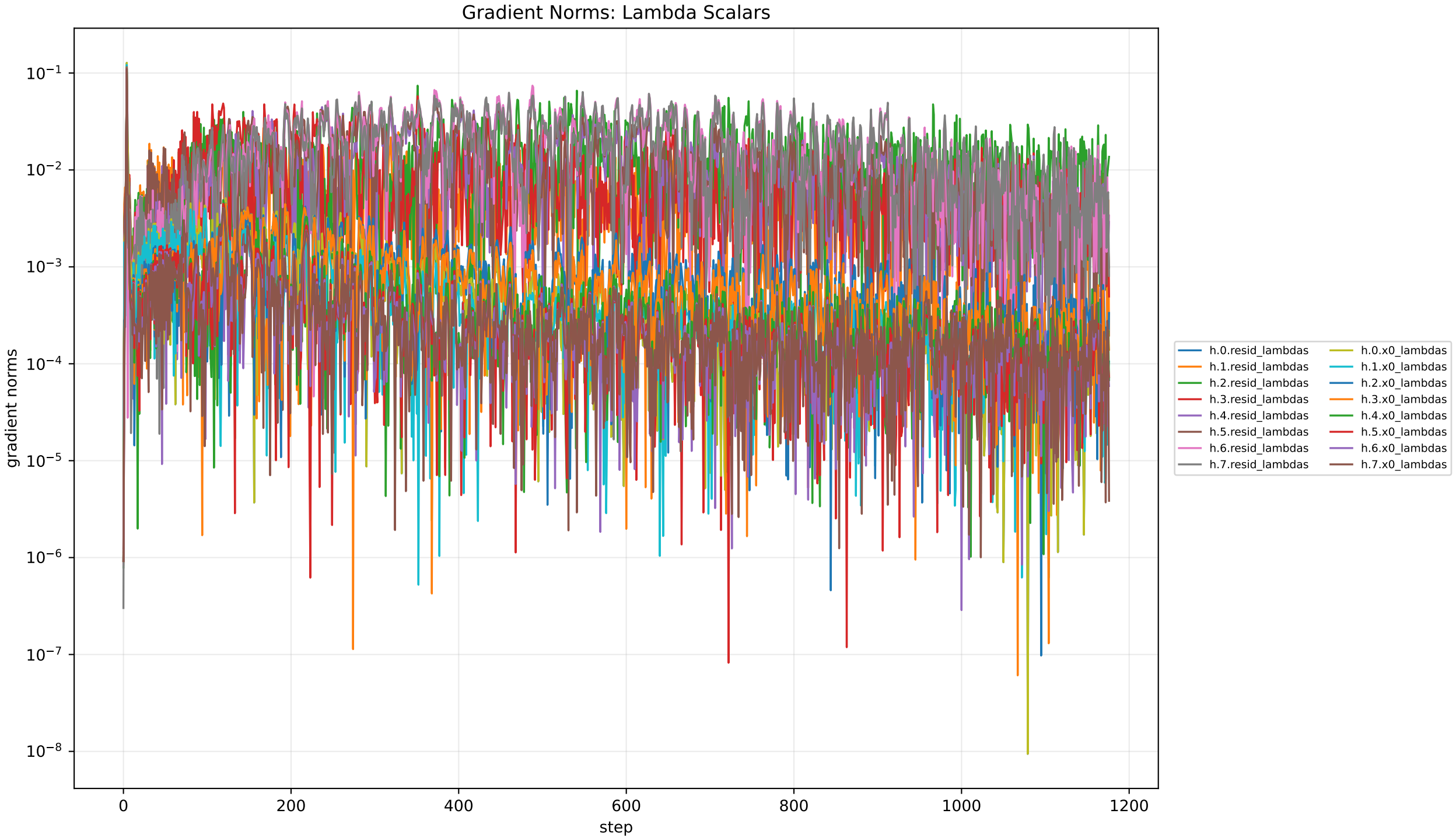

Weight Norms: Lambda Scalars
weight norms
step
10^{1}
10^{0}
10^{-1}
10^{-2}
0
200
400
600
800
1000
1200
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas

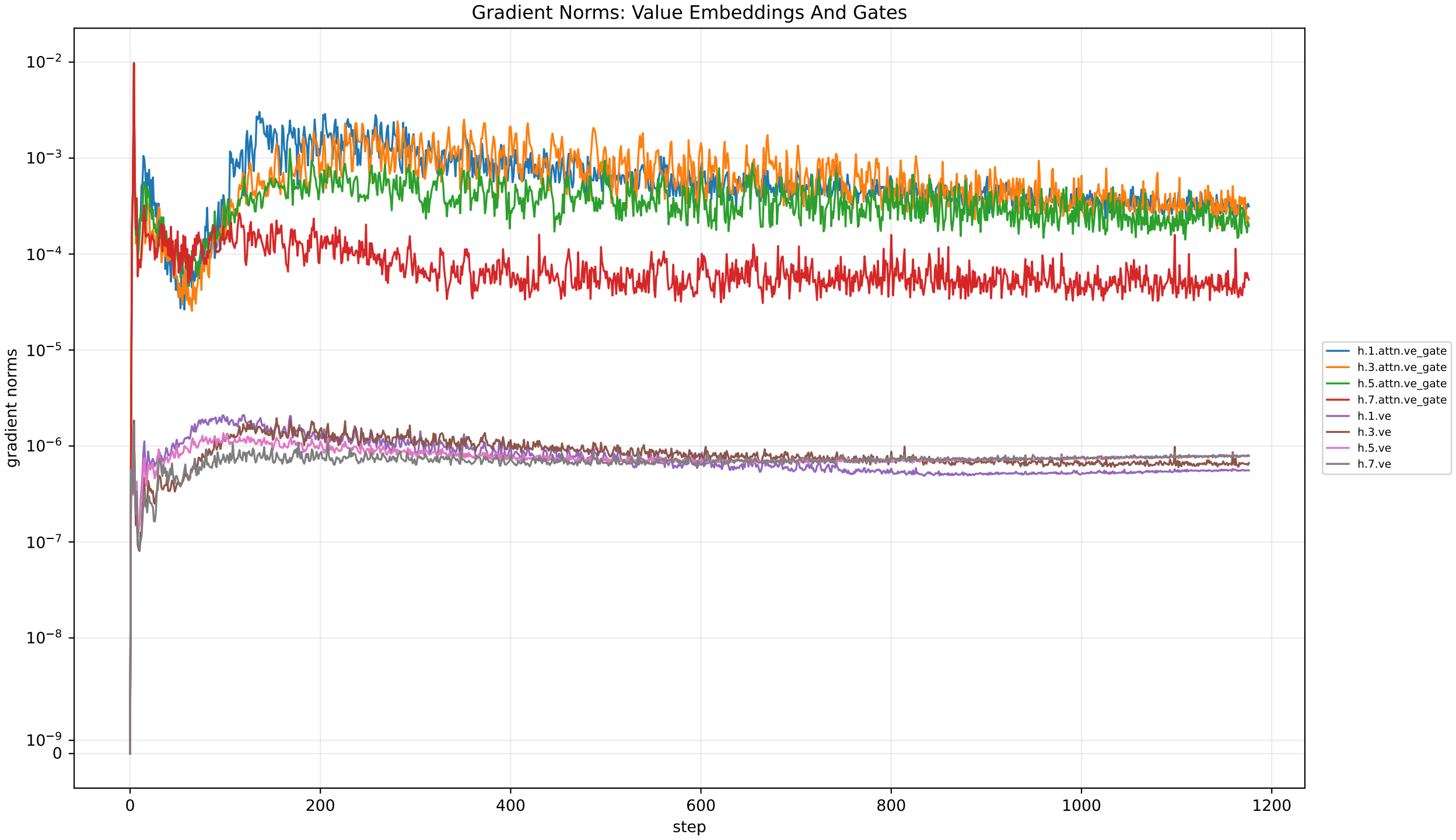

Weight Norms: Value Embeddings And Gates
weight norms
step
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
0
200
400
600
800
1000
1200
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve


Gradient Norms: Embeddings And Head
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
200
400
600
800
1000
1200
lm_head
wte


Gradient Norms: Attention QKV
gradient norms
step
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
0
0
200
400
600
800
1000
1200
h.0.k
h.1.k
h.2.k
h.3.k
h.4.k
h.5.k
h.6.k
h.7.k
h.0.q
h.1.q
h.2.q
h.3.q
h.4.q
h.5.q
h.6.q
h.7.q
h.0.v
h.1.v
h.2.v
h.3.v
h.4.v
h.5.v
h.6.v
h.7.v


Gradient Norms: Attention And MLP Projections
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
0
200
400
600
800
1000
1200
h.0.attn.c_proj
h.1.attn.c_proj
h.2.attn.c_proj
h.3.attn.c_proj
h.4.attn.c_proj
h.5.attn.c_proj
h.6.attn.c_proj
h.7.attn.c_proj
h.0.mlp.c_fc
h.1.mlp.c_fc
h.2.mlp.c_fc
h.3.mlp.c_fc
h.4.mlp.c_fc
h.5.mlp.c_fc
h.6.mlp.c_fc
h.7.mlp.c_fc
h.0.mlp.c_proj
h.1.mlp.c_proj
h.2.mlp.c_proj
h.3.mlp.c_proj
h.4.mlp.c_proj
h.5.mlp.c_proj
h.6.mlp.c_proj
h.7.mlp.c_proj


Gradient Norms: Lambda Scalars
gradient norms
step
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
0
200
400
600
800
1000
1200
h.0.resid_lambdas
h.1.resid_lambdas
h.2.resid_lambdas
h.3.resid_lambdas
h.4.resid_lambdas
h.5.resid_lambdas
h.6.resid_lambdas
h.7.resid_lambdas
h.0.x0_lambdas
h.1.x0_lambdas
h.2.x0_lambdas
h.3.x0_lambdas
h.4.x0_lambdas
h.5.x0_lambdas
h.6.x0_lambdas
h.7.x0_lambdas


Gradient Norms: Value Embeddings And Gates
gradient norms
step
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
0
0
200
400
600
800
1000
1200
h.1.attn.ve_gate
h.3.attn.ve_gate
h.5.attn.ve_gate
h.7.attn.ve_gate
h.1.ve
h.3.ve
h.5.ve
h.7.ve